Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Study to Learn About the Study Medicine Aztreonam-Avibactam (ATM-AVI) in Infants and Newborns Admitted in Hospitals With Bacterial Infection (CHERISH) **Short Title/Acronym:** CHERISH Study **Protocol Number:** C3601010 **NCT Number:** NCT06462235 **Sponsor Name:** Pfizer **Investigational Product:** Aztreonam-Avibactam (ATM-AVI) **Phase of Development:** Phase 2a **Medical Monitor:** Pfizer Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To assess the pharmacokinetics (PK), safety, and tolerability of single and multiple doses of ATM-AVI in hospitalized neonates and infants (from birth to <9 months of age) with suspected or confirmed bacterial infections. * **Secondary Objectives:** To evaluate the clinical and microbiological efficacy of ATM-AVI for the possible treatment of infections caused by Gram-negative bacteria.

2.2 Background & Rationale Gram-negative bacteria can cause serious, life-threatening infections (such as complicated intra-abdominal infections [cIAI], complicated urinary tract infections [cUTI], hospital-acquired pneumonia/ventilator-associated pneumonia [HAP/VAP], bloodstream infections [BSI], and sepsis) that are increasingly difficult to treat due to antibiotic resistance. The investigational product combines aztreonam (ATM) with avibactam (AVI), a β-lactamase inhibitor, to restore ATM's activity against resistant pathogens. This trial aims to establish the proper dosing, safety, and PK profile of this combination for a highly vulnerable pediatric population (under 9 months of age).

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3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm (evaluation across multiple cohorts) * **Blinding:** Open-label * **Randomization:** Non-randomized

3.2 Planned Enrollment & Duration * **Target N:** 48 participants (divided into 4 age cohorts of 12 infants each; 6 in Part A and 6 in Part B) * **Number of Sites:** Multicenter global study * **Estimated Study Start:** 25-Sep-2024 * **Estimated Primary Completion:** 02-Dec-2027

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Hospitalized neonates and infants from birth to <9 months of age, including preterm birth. 2. **Part A:** Receiving IV antibiotics for the treatment of suspected or confirmed bacterial infection, including but not limited to cIAI, cUTI, HAP/VAP, BSI, or sepsis. 3. **Part B:** Suspected or confirmed Gram-negative bacterial infection requiring IV antibiotics, including but not limited to cIAI, cUTI, HAP/VAP, BSI, or sepsis.

4.2 Exclusion Criteria 1. Severe renal impairment or known significant renal disease (evidenced by elevated serum creatinine at screening, urinary output <0.5 mL/kg/h for 6 consecutive hours, or requirement for dialysis). 2. **Part B Only:** Received >24 hours of systemic antibiotic treatment for Gram-negative organisms at the time of enrollment, unless there is a documented treatment failure or lack of improvement in at least one objective sign or symptom after ≥ 48 hours of antibiotics. 3. Any medical or laboratory abnormality that may increase the risk of study participation or make the participant inappropriate for the study, per the investigator's judgment.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** * *Part A:* A single 3-hour intravenous infusion of ATM-AVI. * *Part B:* Multiple 3-hour intravenous infusions of ATM-AVI every 6 hours (every 8 hours for preterm infants). * **Route of Administration:** Intravenous (IV) * **Duration of Treatment:** Part A: Single dose. Part B: 3 to 14 days of treatment.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Participants with cIAI will also receive intravenous metronidazole. All participants have the option to receive other appropriate IV antibiotic treatments for Gram-positive bacteria. * **Prohibited:** Prolonged prior systemic Gram-negative antibiotic therapy (>24 hours) for Part B, unless criteria for treatment failure are met.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -1 to 0 | Informed Consent from parent/guardian, Medical History, Baseline Labs, Renal Function Assessment | | **Baseline & Infusion** | Day 1 | **Part A:** Single 3-hour IV infusion, PK blood sampling during and up to 5 hours post-infusion.
Part B: First 3-hour IV infusion initiation. | | **Treatment (Part B)**| Days 1 to 14 | IV ATM-AVI every 6 (or 8) hours. Total of 5 PK blood assessments over the first ≥ 2 days. | | **End of Treatment** | End of IV Tx | Clinical response assessment at End of Intravenous (EOIV) / End of Treatment (EOT). | | **Test-of-Cure**

(TOC) | 7-14 Days post-EOT| Efficacy and microbiological outcome evaluations. | | **End of Study** | 4-5 Weeks post-dose| Final safety follow-up assessment (may be conducted by telephone). Total study duration up to 7 weeks. |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Definition:** Pharmacokinetic (PK) parameters, including Maximum observed plasma concentration (C_{max}), Area under the curve (AUC), Clearance, and Half-life of ATM and AVI as estimated by population pharmacokinetic (popPK) analysis. Safety endpoints include the percentage of participants experiencing Adverse Events (AEs) and Serious Adverse Events (SAEs). * **Measurement Method:** Serial PK blood sampling on Day 1 and at steady state (Day 2 or later); continuous clinical monitoring for safety and tolerability.

7.2 Secondary & Exploratory Endpoints * **Clinical Response:** Assessment of clinical cure/failure at TOC (7 to 14 days after the last antibiotic treatment). * **Microbiological Outcomes:** Eradication, presumed eradication, persistence, presumed persistence, or indeterminate.

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8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Inpatient hospital monitoring during infusion, continuous observation for 48 hours post-dose (Part A), and through the 4-5 week final safety follow-up (Parts A & B).
- **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for severe and life-threatening events.
- **Data Monitoring Committee (DMC):** An independent external DMC will review plasma drug levels, safety, and tolerability data. For example, enrollment of Part A Cohort 4 (preterm neonates) will only commence after sponsor and DMC review of data from previous cohorts.

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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** PK Analysis Set (for popPK modeling) and Safety Set (all patients who receive any amount of the study drug).
- **Sample Size Justification:** A sample size of 48 participants across 4 distinct age cohorts is determined adequate to characterize the pediatric PK profile and identify preliminary safety trends prior to larger efficacy trials.
- **Handling of Missing Data:** Sparse PK sampling data will be handled using a population PK (popPK) modeling approach.

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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** High-frequency onsite inpatient monitoring during active treatment, supplemented by DMC milestone reviews and a remote telephone follow-up at the study's conclusion.
- **Audit Trail:** Strict documentation of IV preparation, PK sample timing, and clearance evaluations.

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11. Study Identifiers & Governance

- **Primary ID:** C3601010
- **Registry Number:** NCT06462235
- **Sponsor/Lead Organization:** Pfizer
- **Study Title:** CHERISH
- **Official Regulatory Title:** A Study to Learn About the Study Medicine Aztreonam-Avibactam (ATM-AVI) in Infants and Newborns Admitted in Hospitals With Bacterial Infection (CHERISH)

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12. Clinical Framework (Infection Specific)

- **Primary Condition:** Gram-negative Bacterial Infection (e.g., cIAI, cUTI, HAP/VAP, BSI, sepsis)
- **Diagnosis Threshold:** Suspected or confirmed bacterial infection requiring hospital admission and IV antibiotics
- **Medical Methodology:** Antimicrobial therapy
- **Optimization Target:** Identifying safe and effective dosages of ATM-AVI for infants <9 months to combat antibiotic resistance

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13. Study Procedures & Methodology

- **Management Pathway:** Inpatient intensive care and continuous IV antibiotic administration
- **Education Protocol (HCP):** Extensive site training on weight/age-based dosing, exact PK sampling timelines, and safe IV infusion
- **Education Protocol (Patient):** Informed consent processes and risk/benefit counseling for parents/legal guardians
- **Quality Audit Mechanism:** Periodic independent DMC reviews before unlocking progressive age cohorts

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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 7 weeks per participant
- **Onsite Visit Cadence:** Daily inpatient monitoring throughout the 3–14 day treatment cycle and for 48 hours post-treatment
- **Remote Monitoring Frequency:** Final follow-up phone call at 4–5 weeks post-dose
- **Checklist Review Interval:** Routine hospital vitals and laboratory checks continuously during inpatient stay.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]